Ch 2.1: What is Statistical Learning?

Lecture 2 - CMSE 381

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::
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Weds, Aug 28, 2024

Announcements

Last time:

- Discussed where to find everything
  - Course webpage
  - Slack
  - D2L
- Check out the syllabus!

Lec #	Date			Reading	HW
1	Mon	8/26	Intro / First day stuff / Python Review Pt 1	1	
2	Wed	8/28	What is statistical learning?	2.1	
	Fri	8/30	Class Cancelled (Dr Munch out of town)		
	Mon	9/2	No class - Labor day		
3	Wed	9/4	Assessing Model Accuracy	2.2.1, 2.2.2	
4	Fri	9/6	Linear Regression	3.1	HW #1 Due

cmse.msu.edu / CMSE 381

Announcements:

- Get on slack!
  - +1 point on the first homework if you post a gif in the thread
- First homework due Sun Sep 8
- First office hours next week

Covered in this class

- Input/output variables
- Prediction vs inference
- Reduceable vs irreduceable error
- Overfitting
- Classification vs regression
- Supervised vs Unsupervised learning

- Please note: no jupyter notebook for today's class, slides only

An example data set: `Advertising`

	TV	Radio	Newspaper	Sales
1	230.1	37.8	69.2	22.1
2	44.5	39.3	45.1	10.4
3	17.2	45.9	69.3	9.3
4	151.5	41.3	58.5	18.5
5	180.8	10.8	58.4	12.9
6	8.7	48.9	75	7.2
7	57.5	32.8	23.5	11.8
8	120.2	19.6	11.6	13.2
9	8.6	2.1	1	4.8
10	199.8	2.6	21.2	10.6
11	66.1	5.8	24.2	8.6

Fake fake

- Sales of a product in 200 markets, along with amount spent on three differnt types of advertising
- Goal: Predict sales based on the
- Input variables: amount spent on each
- Output variable: type of advertising

Input : TV, Radio, Newspaper

Output: Sales

Data available at msu-cmse-courses.github.io/CMSE381-F24/DataSets/DataSets.html

Notation and Big Assumption

Input variables: $X_1, X_2, \cdots, X_p$

Output variable: Y

# of input variables $= p$

Ground truth
Want but don't have
Systemic information that X provides about Y

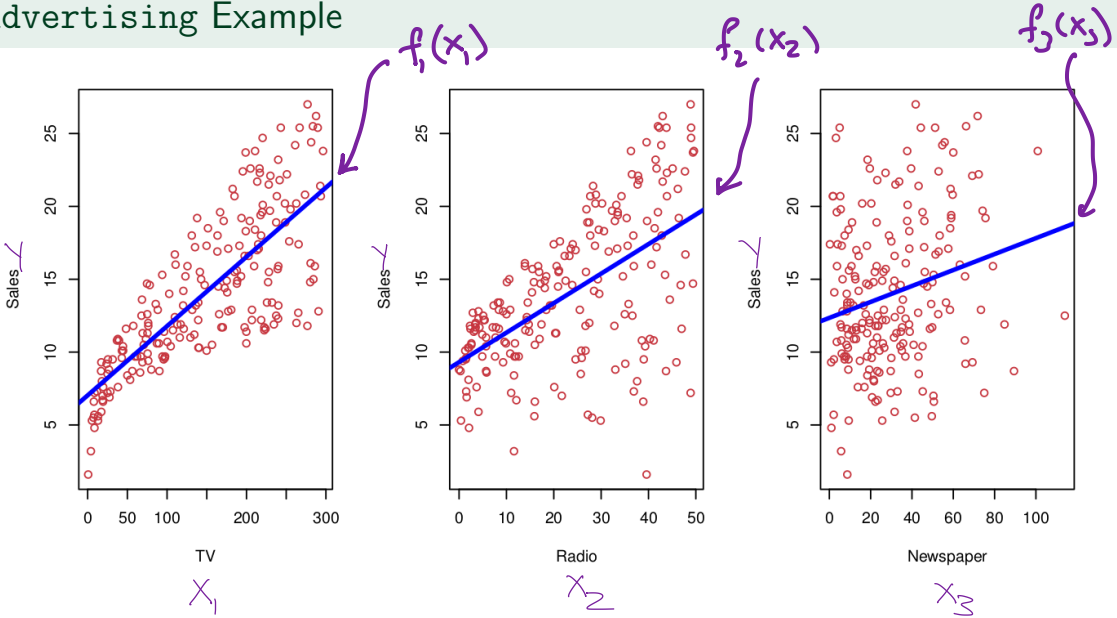
$$Y = f(X) + \varepsilon$$

$f(X_1, X_2, \ldots, X_p)$

Goal: Come up with an estimation of f, denoted $\hat{f}$

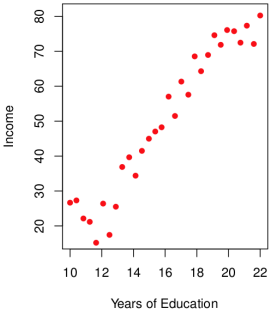
ε: a random error term which is independent of X and has mean 0.

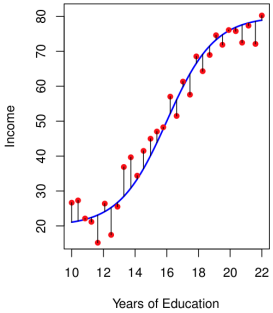
Advertising Example

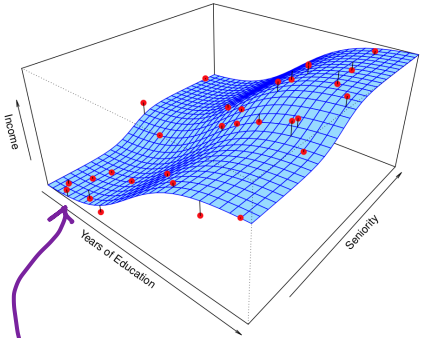


More examples

f(years of ed) = income







f(yrs ed, seniority)
= income

Section 1

Prediction vs Inference

Prediction

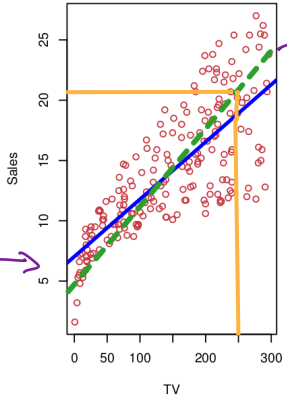
Example: If we spend \$250 on TV advertising, what do we predict we will we make in sales?

Given a value X, try to provide an estimate for $f(X)$.

Build a model:

$$\hat{Y} = \hat{f}(X)$$

Want a good guess for f which is unknown



blue

$\hat{f}$

green dashed approximation

$\hat{f}(250) \approx \$21$ sales

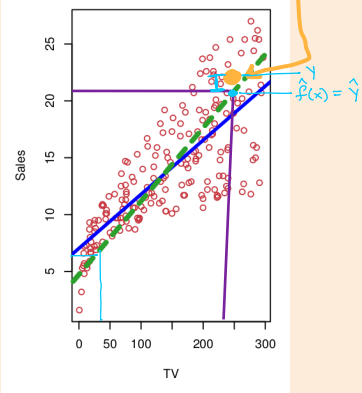
Group question:

	TV	Radio	Newspaper	Sales
1	230.1	37.8	69.2	22.1
2	44.5	39.3	45.1	10.4
3	17.2	45.9	69.3	9.3

$\hat{f}(x)$ 21 7 5

Err 1.1 3.4 4.3

$y - \hat{y} = \cancel{f(x)} - \hat{f}(x)$



The blue solid line is f. The green dashed line is $\hat{f}$.

- What is the predicted sales for the first three data points using the green dashed line $\hat{f}$ shown in the graph?
- Using the dashed green line as the predicted model $\hat{f}$, what is the error in each of the three predicions?

Reduceable vs irreducable error

All models are wrong, some are useful.

$$Y - \hat{Y}$$

Reducible Error

- $\hat{f}$ will not be a perfect estimate for f
- We can potentially improve reducable error by getting a better model

Irreducible Error

- $Y = f(x) + \varepsilon$
- Variability of ε affects our predicting
- No matter how well I do predicting f can't get rid of this error

More on error

- Given estimate $\hat{f}$ (fixed)
- Set of predictors X (fixed)
- Prediction $\hat{Y} = \hat{f}(X)$

$$E(\underbrace{Y - \hat{Y}}_{\text{error}})^2 = E\left[f(x) + \varepsilon - \hat{f}(x)\right]^2$$

$$= \underbrace{\left[f(x) - \hat{f}(x)\right]^2}_{\text{reducable error}} + \underbrace{Var(\varepsilon)}_{\text{irreducable error}}$$

Inference

Want f, but not for prediction (or possibly combined with prediction)

- Which predictors are associated with the response?
- What is the relationship between the response and each predictor?
- Can the relationship between Y and each predictor be adequately summarized using a linear equation? Is it more complicated?

Determine whether each scenario is prediction, inference, or both.

Application	Prediction	Inference
Predict effectiveness of vaccine	X	
Determine the address written on the image of an envelope.	X	
Identify risk factors for getting long covid.		X
Transcribe an audio file of a person talking.	X	
Predict stock prices.	X	

Section 2

How to estimate f?

Input: Training data

- n data points observed
- x_{ij} is the jth predictor for observation i
- y_i is the response variable for the ith observation
- Training data:
  - $\{(x_1, y_1), (x_2, y_2), \cdots, (x_n, y_n)$
  - $x_i = (x_i, x_{i2}, \cdots, x_{ip})^T$

p = # of input variables

$(x_1, y_1) = ((x_{11}\ x_{12}\ x_{13} \ldots x_{1p}), y_1)$

X_{s2}

2nd predictor

# of rows = n

5th observation

ys

		TV	Radio	Newspaper	Sales
1		230.1	37.8	69.2	22.1
2	2	44.5	39.3	45.1	10.4
3	3	17.2	45.9	69.3	9.3
4	4	151.5	41.3	58.5	18.5
5	5	180.8	10.8	58.4	12.9
6	6	8.7	48.9	75	7.2
7	7	57.5	32.8	23.5	11.8
8	8	120.2	19.6	11.6	13.2
9	9	8.6	2.1	1	4.8
10	10	199.8	2.6	21.2	10.6
11	11	66.1	5.8	24.2	8.6

Parametric methods

Step 1: Select a model

Example:

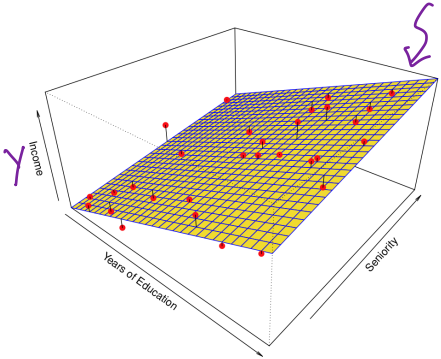
coefficients

$$f(X) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p$$

Step 2: Train the model

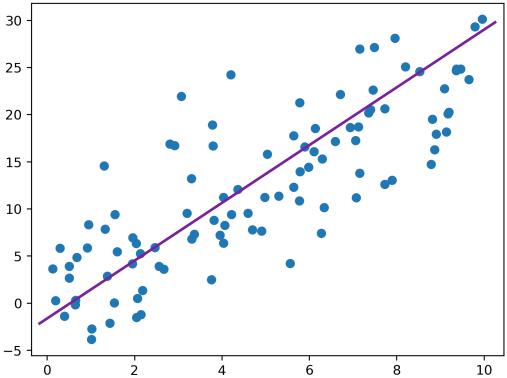
Example:
Find $\beta_i's$ so that

$$Y \approx \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p$$



How do you decide on the coefficients?

$$Y \approx \beta_0 + \beta_1 X_1$$



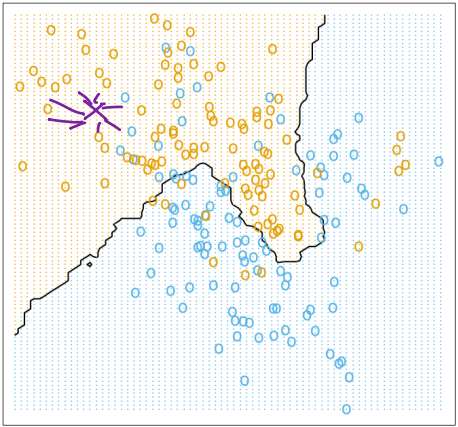
Desmos toy: https://www.desmos.com/calculator/skvt8c73l7

Example Non-parametric method: Nearest Neighbors

$$N_k(x) = \text{Set of } k \text{ nearest neighbors of } x$$

$$\hat{f}(x) = \frac{1}{k} \sum_{x_i \in N_k(x)} y_i$$

X_2



$k = 15$ X_1

Parametric methods: Pros and Cons

Pros

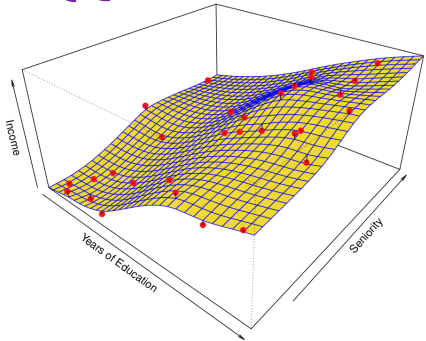
Easier to estimate parameters than to figure out a arbitrary function

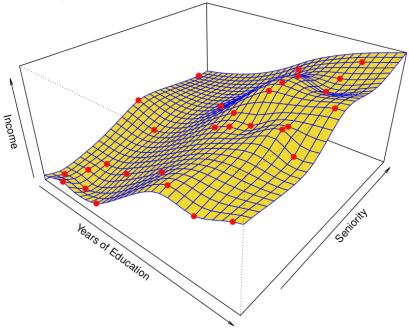
Cons

You might have chosen a wrong function type so the right answer isn't available

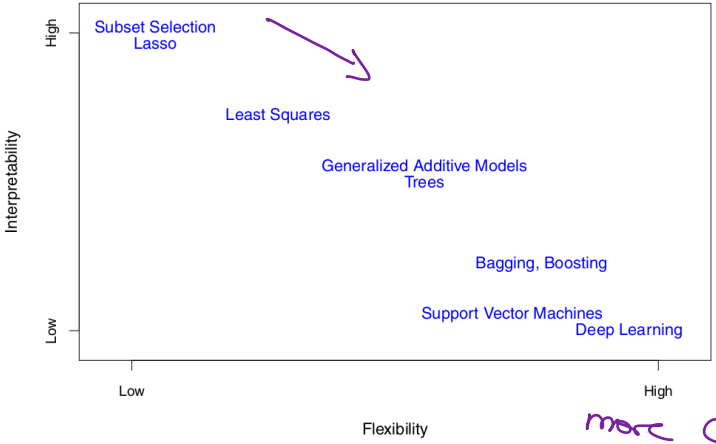
Overfitting – Follow the noise of model, not the model itself

Fix parameterized problem by getting more flexibility (ie. more ~~input variables~~ parameters)



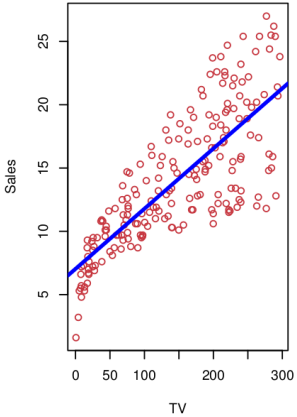


Prediction Accuracy vs Model Interpretability

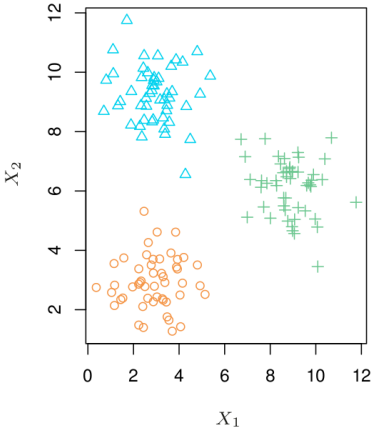


more chance of overfitting

Supervised learning:
Training data has response variable y for every input x



Unsupervised Learning:
Training data does not have response variable y for every input x



Regression vs Classification

Types of variables: Output y

- Quantitative

  Values are continuous numbers (ordered)

  Blood pressure
  height age volume dist

  Regression

- Qualitative / Categorical

  Set of values where order doesn't matter

  yes/no blood type
  job

  Classification

Section 3

Group work

(a) We collect a set of data on the top 500 firms in the US. For each firm we record profit, number of employees, industry and the CEO salary. We are interested in understanding which factors affect CEO salary.

- Is this classification or regression?
- Do we want inference or prediction?
- What is n, the number of data points?
- What is p, the number of variables?

(b) We are considering launching a new product and wish to know whether it will be a success or a failure. We collect data on 20 similar products that were previously launched. For each product we have recorded whether it was a success or failure, price charged for the product, marketing budget, competition price, and ten other variables.

- Is this classification or regression?
- Do we want inference or prediction?
- What is n, the number of data points?
- What is p, the number of variables?

TL;DR

Wrap up

Next time:

- Friday 8/30 and Monday 9/2
  - NO CLASS!!!!
- Wednesday 9/4
  - Bring Laptop!
  - First homework due Sun Sep 8
  - There will be a quiz next week

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5	Mon	9/9	More Linear Regression	3.1/3.2	

Announcements:

- Get on slack!
  - +1 point on the first homework if you post a gif in the thread
- Office hours!
  - Dr. Munch: TBD
  - Christy Lu: TBD